# Designing a Universal Information Language

Oscar Bender-Stone

March 26, 2026

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# Agenda

**Introduction**

Language Overview

Syntax

Metatheory

Conclusion

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# Background

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- Information Management: studies how to **store** and **organize** data

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# Problems

- How to *actually* manage information?
  - **Information base:** a database of information
- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]

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- **Depth:** domain specific data
  - Example: Amazon (product suggestions)[1]
- **Breadth:** different kinds of data
  - Example: Wikipedia[2]
- **Standardization:** ambiguous interpretations
  - Related issue: being **interoperable**
  - **Interoperable:** a portable data format
  - Example: US health records[3]

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# Existing Solutions

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- **Resource Description Language (RDF)**
  - ▸ Used in Web Ontology Language **(OWL)**
  - \+ Built to be **interoperable**
  - – No built-in logic engine
- **Labeled Property Graphs (LPGs)**
  - ▸ Stores **properties** (node/edge metadata)
  - \+ Efficient queries
  - – Not standardized
- **Cyc**
  - ▸ Largest **knowledge base** to date
  - \+ Sophisticated logic engine
  - – Hard-coded, proprietary rules

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# Proposed Solution

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- **Welkin**: an information language

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    - Support: future optimizations
    - Describe improvements *in* Welkin

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  - **Standardization:** be completely *unambiguous*
    - Have a formal specification

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# General Requirements

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# General Requirements

1. Express any computer program

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   - Established notion: **Turing machines**[1]
2. Enable user-defined *meaning*

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- Achieve both through **representations**

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- Track *faithfulness* of representations through **truth**

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  - Written with braces $\{a, b, \ldots, z\}$

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- Representations are units
- Each unit has its own **context**

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# Information

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# Turing Completeness

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# Overview

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# Encoding

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# Invertible Syntax Descriptions

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# The Welkin Grammar

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# Outline of Unambiguity

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